METRO

IT and Japanese Language Center

Internet Café Management System

Project Document

Version 1.0

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Developed by [15th G, Group-2]

Project Members

No.	Name	Role	Responsibility
1.	Zin Moe Hein	Leader	Backend/Database/Testing
2.	Nine Ko Ko	Member	PowerPoint/Documentation
3.	Phone Myint Thu	Member	Front End
4.	Shin Khant Phyo Wai	Member	Back End/Documentation
5.	Win Lin Hein	Member	Front End/Testing
6.	Wai Lin Htun	Member	Database/PowerPoint

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ABSTRACT

Our **Internet Cafe Management System** is a desktop-based JAVA Application built to automate menial and mundane tasks such as manual session tracking, data logging and calculating fees. The System uses RMI Technology to dynamically check between admin and members which make operations smoother and faster than usual. Equipped with a built-in small-scale POS System to make ordering foods and snacks for a member convenient. Also, by using our system overall efficiency of the whole business and data accuracy will improve.

Project Documentation: Internet Café Management System

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CHAPTER 1
INTRODUCTION

1.1 Overview

Internet Café Management System is a JAVA based Desktop application created to support, supervise and manage a modern Internet Café. The System provides tools for both customers (members) and administrators, automating common tasks such as manual session tracking, time extension requests and calculating fees. The system is designed with the intention of replacing traditional manual logs and subpar computer management with a more efficient, and real-time client-server approach.

This project aimed to create a seamless experience for both members and administrators. From a clean, fresh, modern UI for the members and strong, reliable real-time control panel on the admin side. The system also ensures effective communication between client and admin using RMI technology. Not only does it enhance the user's experience, it also reduces the workload of admin by automating common but mindless tasks such as session and request handling.

1.2 Advantages and Disadvantages

1.2.1 Advantages

1.Real-Time Synchronization

Utilizes RMI-based (Remote Method Invocation) communication to keep the admin panel, server, and client terminals instantly synchronized. Critical metrics—such as active timers, user point balances, and PC statuses—update every second without requiring manual refreshes.

2. Automated Session Lifecycle Management

Handles the entire lifecycle of a user session autonomously. The system manages user authentication, live point consumption, automated logouts upon reaching a zero balance, and precise cost tracking without requiring manual administrator intervention.

3. Integrated Point-of-Sale (POS) & Inventory Sync

Features an embedded food ordering system that allows members to browse items, manage a cart, and place orders. The application automatically updates inventory levels, deducts user points, and logs transactions directly into the active session history for a seamless user experience.

4.Optimized Resource & Memory Management

Engineered with strict cleanup protocols for callbacks, background threads, and caches. This prevents memory leaks and ensures long-term application stability during extended operational periods.

1.2.2 Disadvantages

1.Volatile Notification Storage

Administrative notifications are retained strictly in-memory. If the admin application restarts, all historical alerts are lost, preventing administrators from conducting retrospective reviews or audits of past events.

2.Minimal External Security Hardening

Because the system was designed primarily for trusted, internal network environments, it currently lacks robust defensive measures against external security threats or unexpected user input errors.

3.Database Scalability Bottlenecks

The architecture relies on a single shared database connection across all modules. While synchronized, this tightly coupled design limits horizontal scaling (e.g., running multiple admin instances) and poses a potential performance bottleneck under heavy concurrent loads.

4.Hard-Coded Client Identifiers

Workstation names within the client application are currently hard-coded (e.g., "pc-1"). This heavily restricts deployment flexibility, as multi-station environments require individual builds or manual configuration changes.

CHAPTER 2
METHODOLOGY

2.1 Prototype Model

The development of the Internet Cafe Management System adopted the Evolutionary Prototype Model, an iterative software development approach particularly suited for projects with evolving requirements and complex user interactions. This methodology was selected based on several critical factors:

1. Requirement Volatility: Construction management processes exhibit variability across organizations, necessitating flexible requirement accommodation.
2. User Experience Criticality: The system's success fundamentally depends on intuitive interfaces for non-technical construction professionals.
3. Algorithm Validation: EVM calculations and progress tracking algorithms required empirical validation through successive refinement cycles.
4. Stakeholder Engagement: Continuous involvement of domain experts (project managers, site engineers) ensured practical relevance and usability.

- Phase 1 (Proof-of-Concept): Database architecture and validation of core CRUD operations.
- Phase 2 (Functional Prototype): Implementation of critical workflows, including user authentication and RMI network configuration
- Phase 3 (Enhanced Prototype): Development and integration of foundational methods for both client and server applications.
- Phase 4 (Production Prototype): Performance optimization and seamless data synchronization between the client and server.
- Phase 5 (Deployment Prototype): Finalization of installation packaging and completion of system documentation.

2.2 Technical and Network Errors

1. RMI Registry Initialization Failure

- **Problem Statement:** The system periodically fails to launch or connect to the server due to an RMI (Remote Method Invocation) Registry failure.
- **Root Cause:** The server either fails to start properly during initialization or the designated RMI ports are being actively blocked by the host machine's firewall. Additionally, "zombie" processes from previous crashed sessions may be holding the port captive.
- **System Impact:** Total system block. Admin and member applications cannot communicate, preventing user login and session tracking.

2. Unstable Database Connectivity (Aiven Cloud)

- **Problem Statement:** The application drops connections to the remote Aiven Cloud Database, leading to failed queries and system timeouts.
- **Root Cause:** The architecture relies on a persistent connection to a remote cloud database. Due to local network instability and fluctuating latency, the TCP/IP connection drops, causing JDBC connection timeouts.
- **System Impact:** Data loss, failed session saves, and inability to retrieve member credentials.

3. Asynchronous UI Thread Freezing

- **Problem Statement:** The application’s user interface becomes completely unresponsive, which subsequently triggers critical point calculation errors.
- **Root Cause:** Network tasks (like fetching data from Aiven Cloud or communicating with the RMI server) are likely blocking the main UI thread. When the unstable internet causes a network delay, the UI freezes.
- **System Impact:** Because the session timer and point calculation logic are tied to UI updates rather than a background service, a frozen UI causes the internal clock to drift. This results in inaccurate point deductions for active members.

2.3 Operational and Logic Errors

2.3.1 Point Calculation & Deduction Mismatch

- **Problem Statement:** The system exhibits a logic flaw when deducting points for time used, specifically failing to account for session times under the 6-minute threshold.
- **Root Cause:** The current business logic states that 1 point is deducted forever 6 minutes of use. However, the system fails to parse fractions of this time block. Attempts to patch the logic to account for sub-6-minute sessions have resulted in state synchronization issues, where the Admin dashboard and the Member client display completely different point balances.
- **System Impact:** Financial/Operational discrepancy. Users either get free time (under 6 minutes) or the system displays conflicting data, eroding trust in the billing mechanism.

2.3.2 Client-Side Timer Desynchronization

- **Problem Statement:** The active session timer on the client side behaves erratically, occasionally freezing, slowing down, or speeding up.
- **Root Cause:** The timer relies heavily on real-time network stability to sync with the server. When packet loss occurs or latency spikes, the sync fails, causing the timer to stall or overcompensate when the connection is re-established.
- **System Impact:** Inaccurate tracking of user sessions, directly tying back to the point deduction errors mentioned above.

CHAPTER 3

SYSTEM DESIGN

3.1 Use Case Diagram

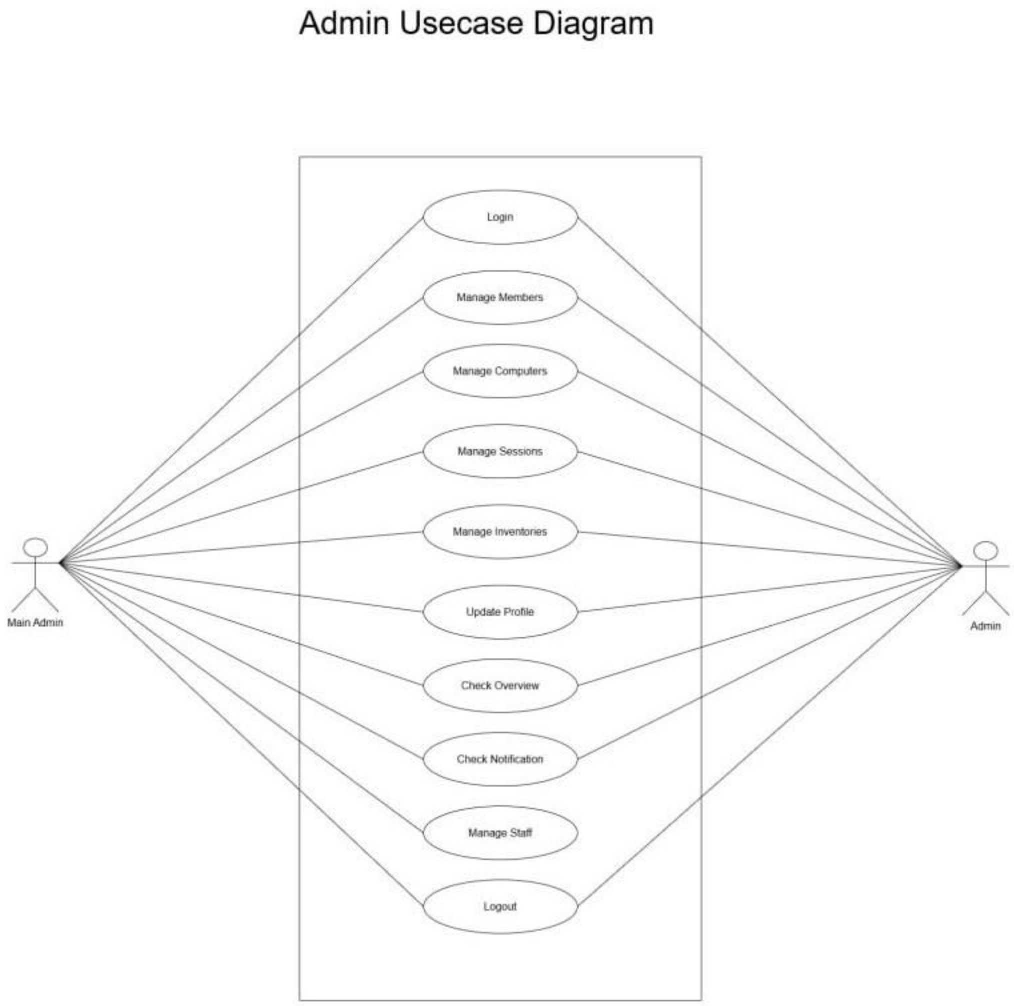

Figure 3.1 Admin Use Case Diagram

3.2 Member Use Case Diagram

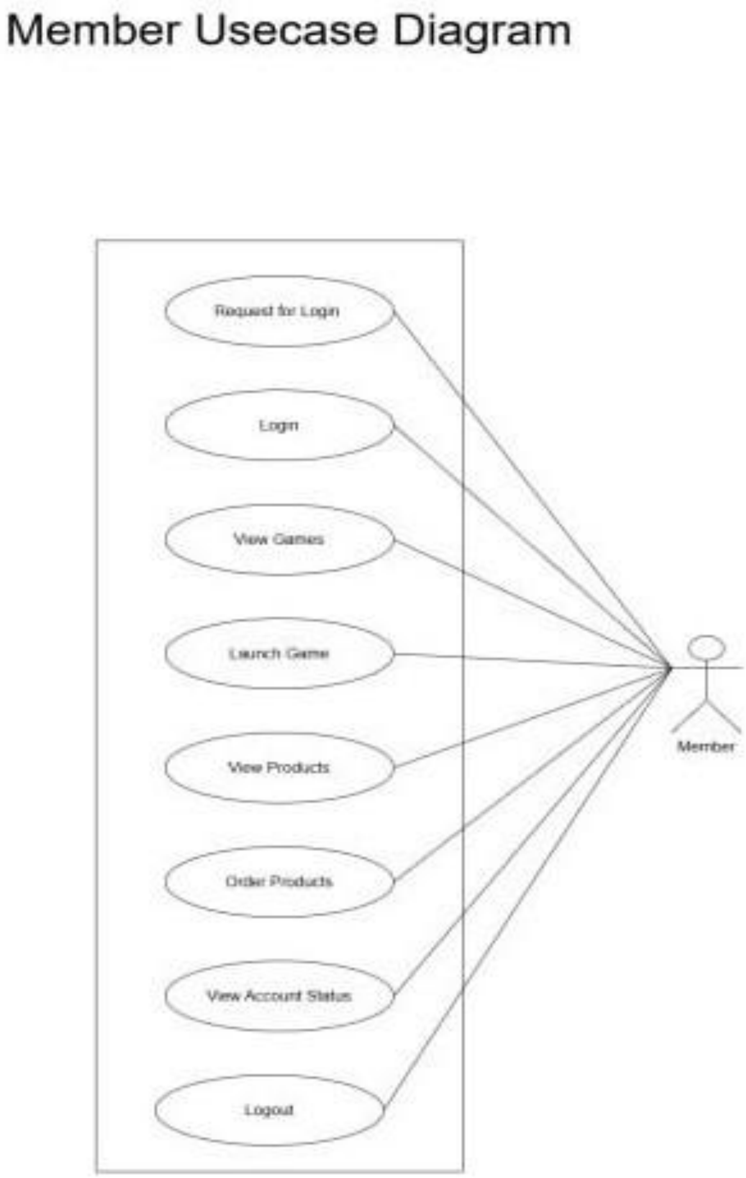

Figure 3.2 Member Use Case Diagram

3.3 Admin Flow Chart

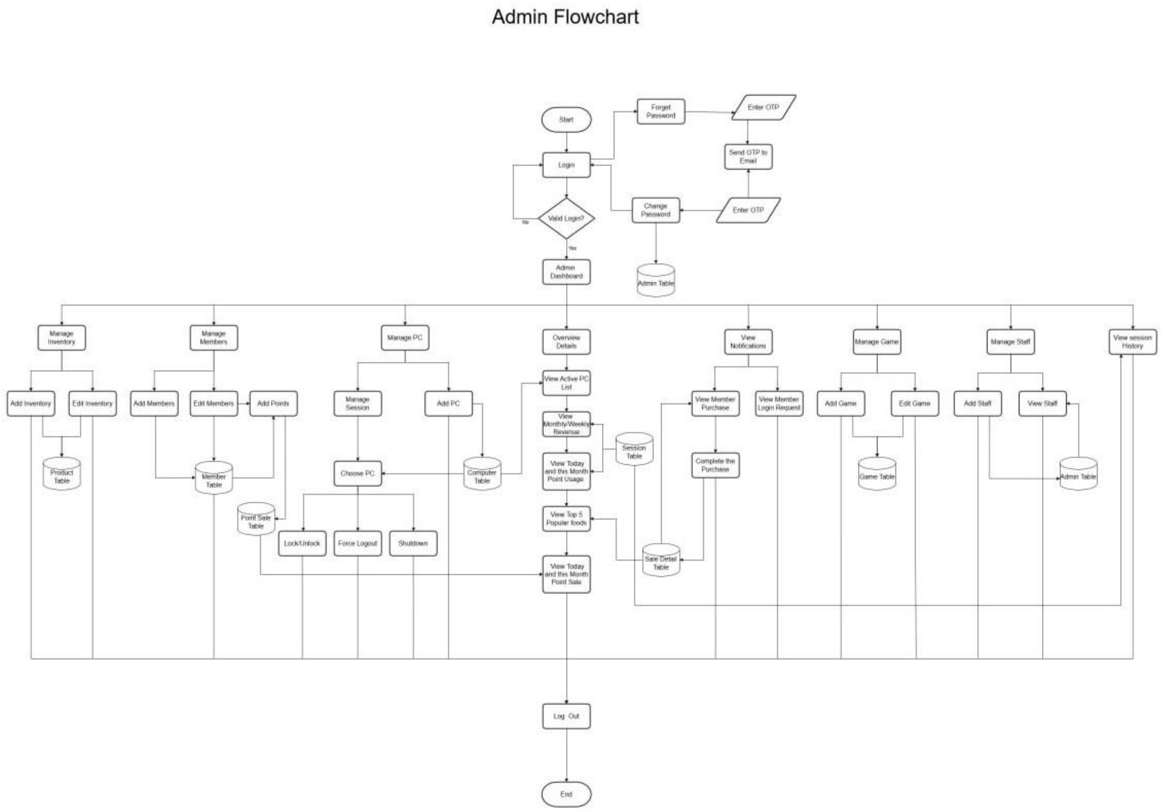

Figure 3.3 Admin Flow Chart

3.4 Member Flow Chart

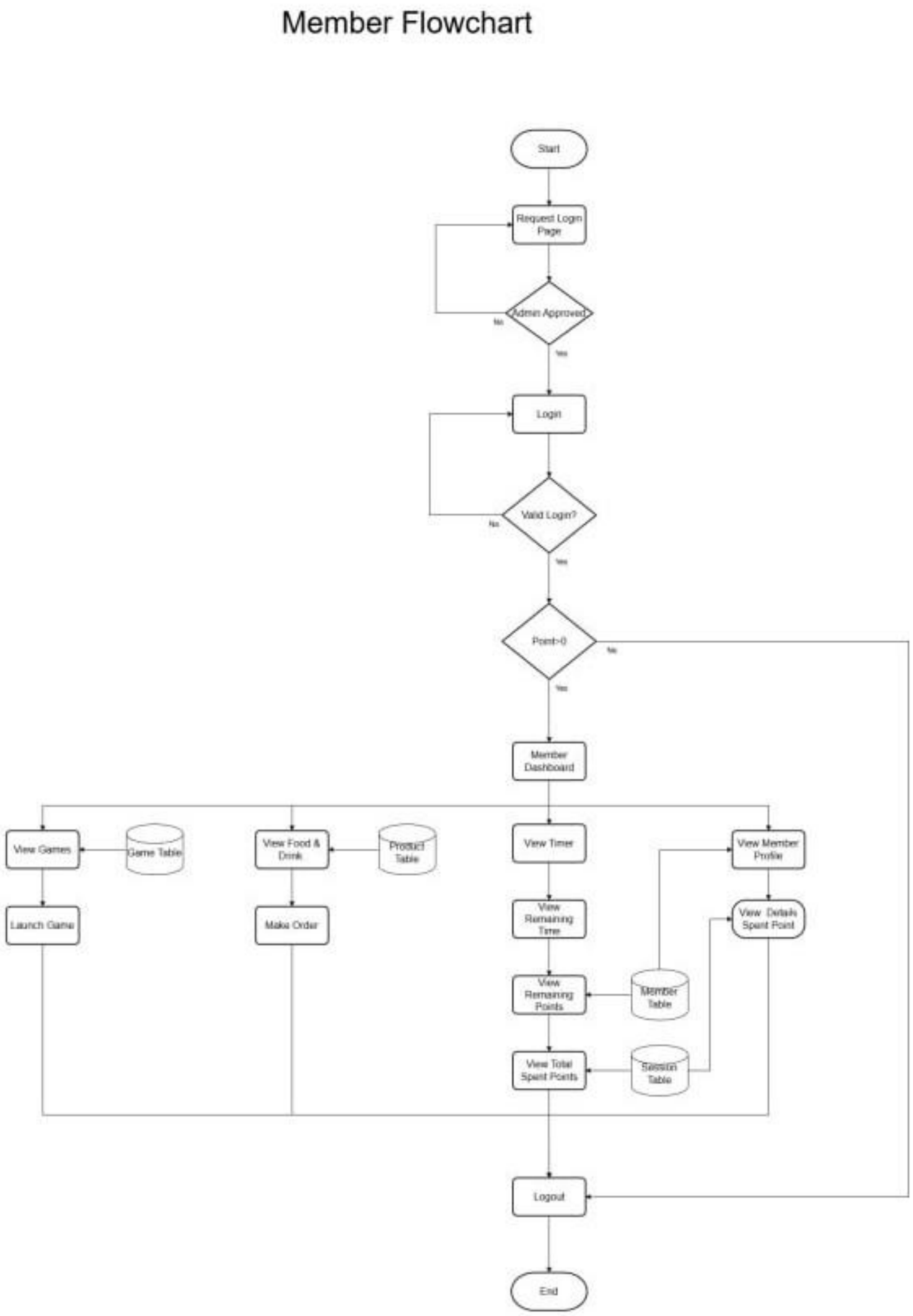

Figure 3.4 Member Flow Chart

CHAPTER 4

IMPLEMENTATION

4.1Technologies Used

Tool Category	Specific Tools	Purpose
Programming Language	Java (JDK 21)	Core Application Development
Database	MySQL 8.0	Data Storage and Management
GUI Design	JavaFX, Scene Builder	User Interface
IDE	Apache NetBeans	Integrated Development Environment
Libraries	FontAwesomeFx, Jakarta.Activation, Javax.Activation, Javax.mail	Icons, Data Manipulation, Identification, SMTP
Styling	CSS	User Interface Styling

4.2 Database Design

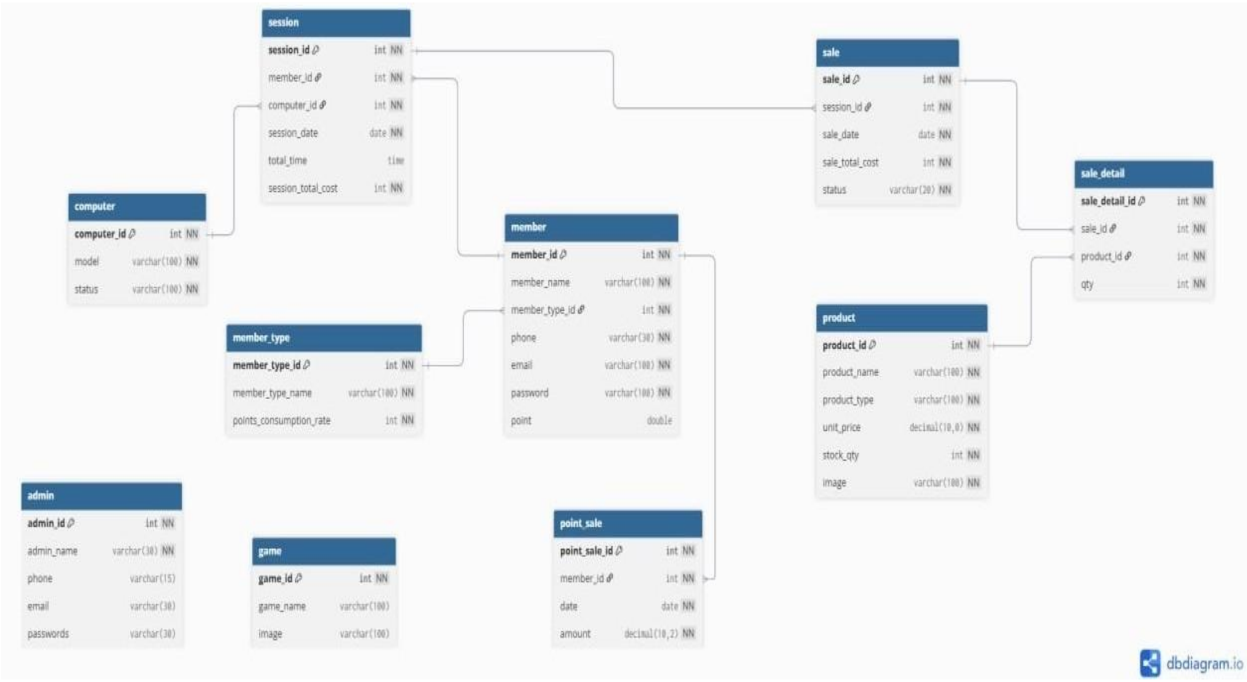

Figure 4.2 Database Design

CHAPTER 5

USER INTERFACE

5.1 Admin Interface

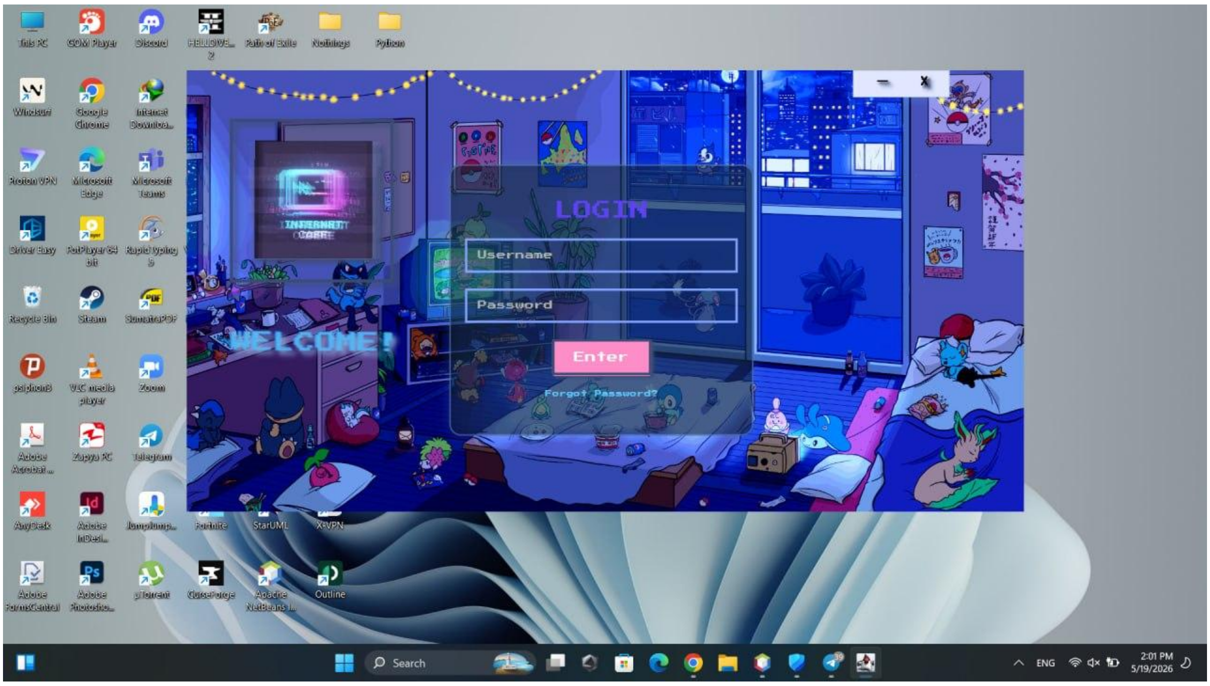

Figure 5.1 Admin's Login Interface

Figure 5.2 Admin's Dashboard

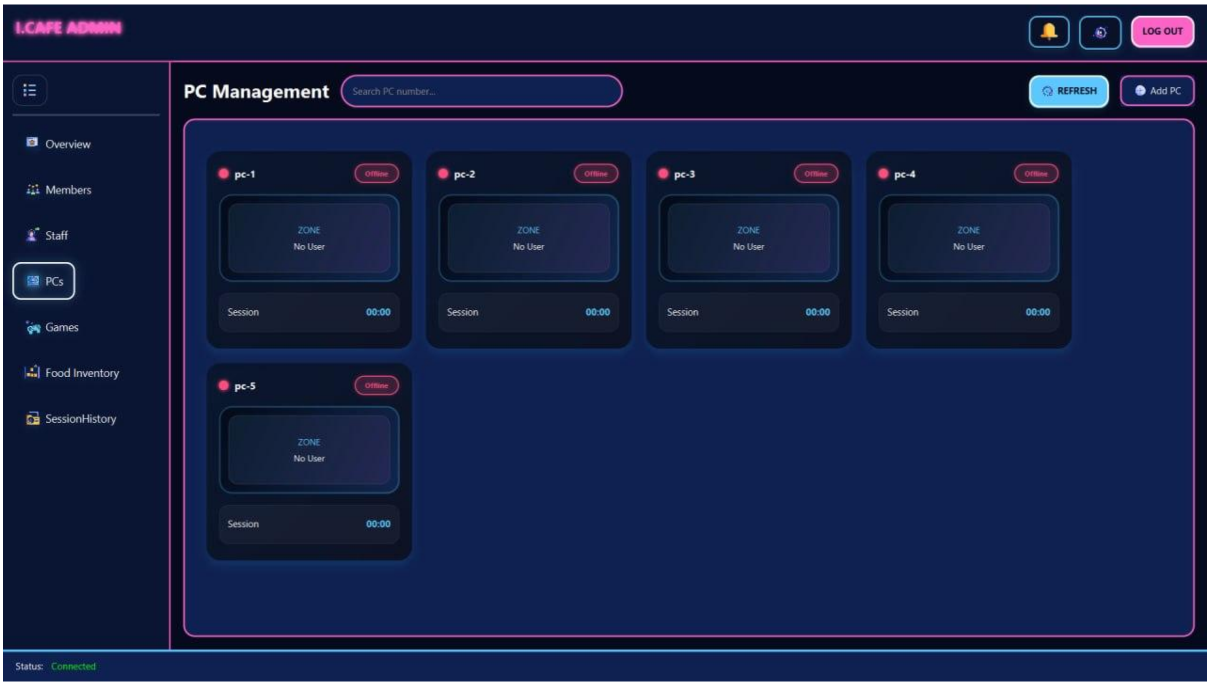

Figure 5.3 PC View Controller

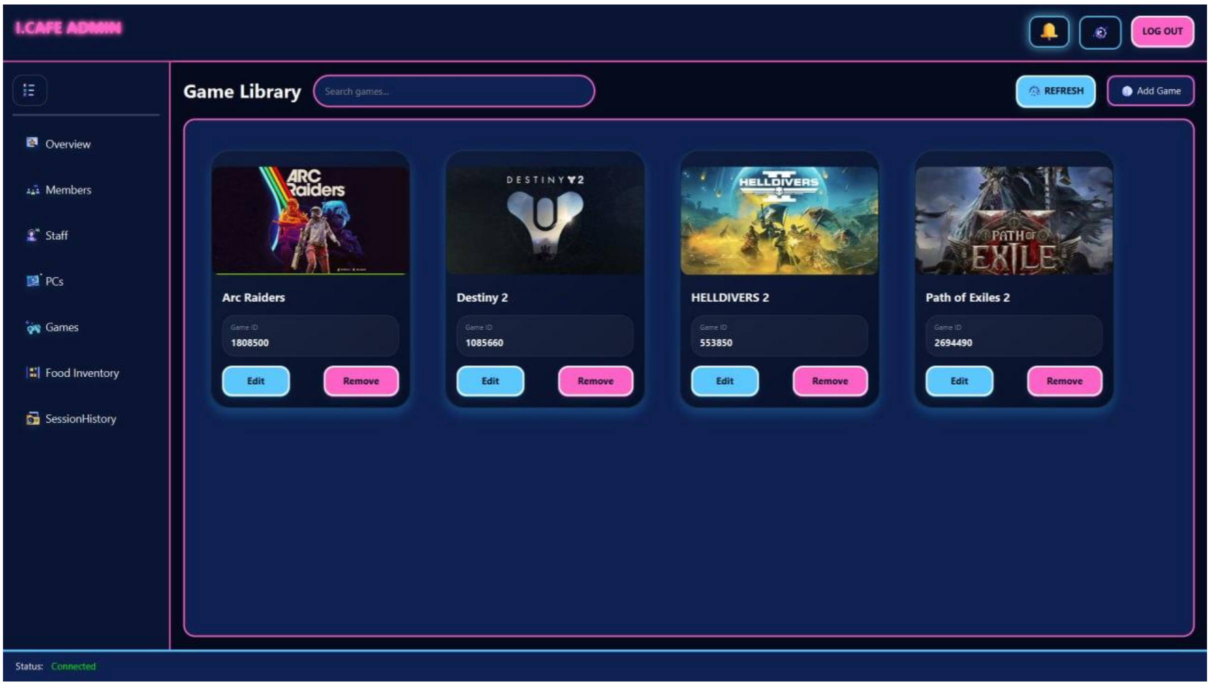

Figure 5.4 Game Controller

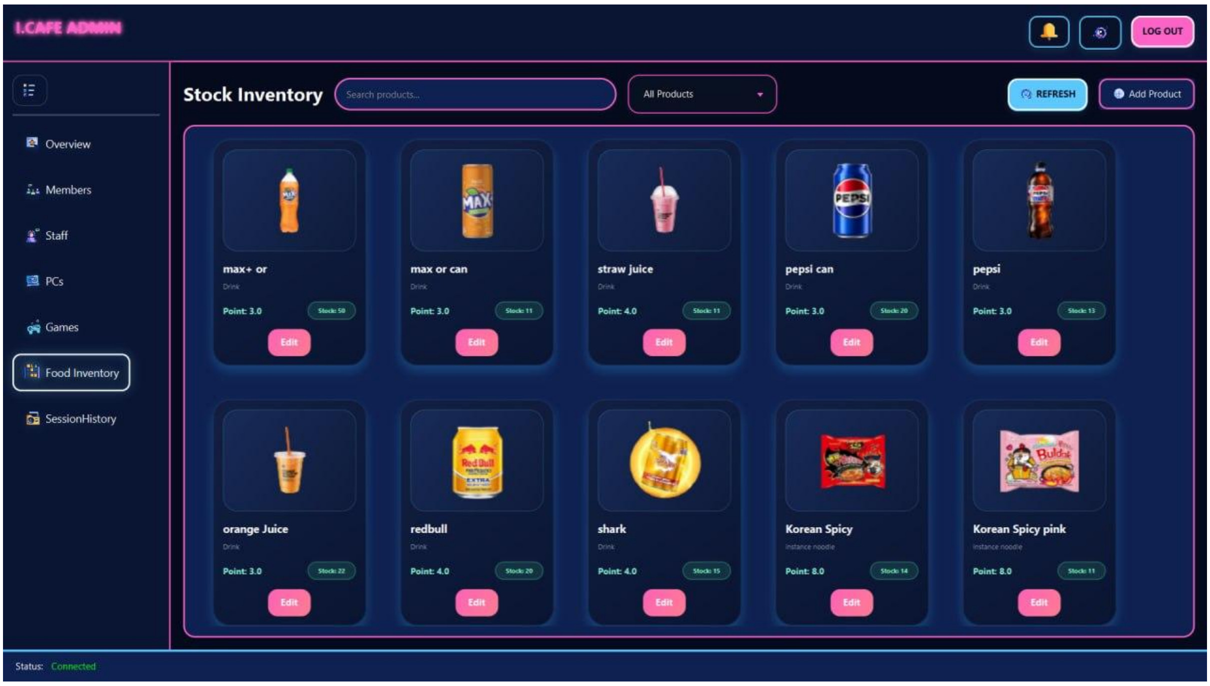

Figure 5.5 Inventory Interface

5.2 Member Interface

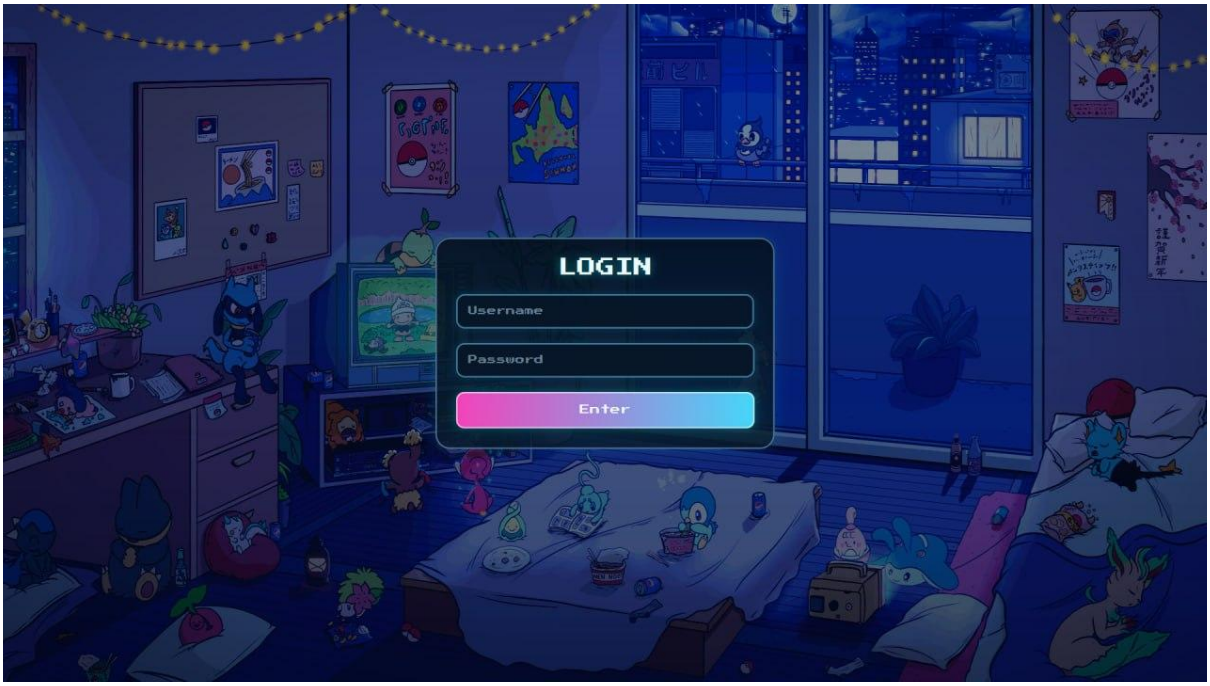

Figure 5.6 Member's Login Interface

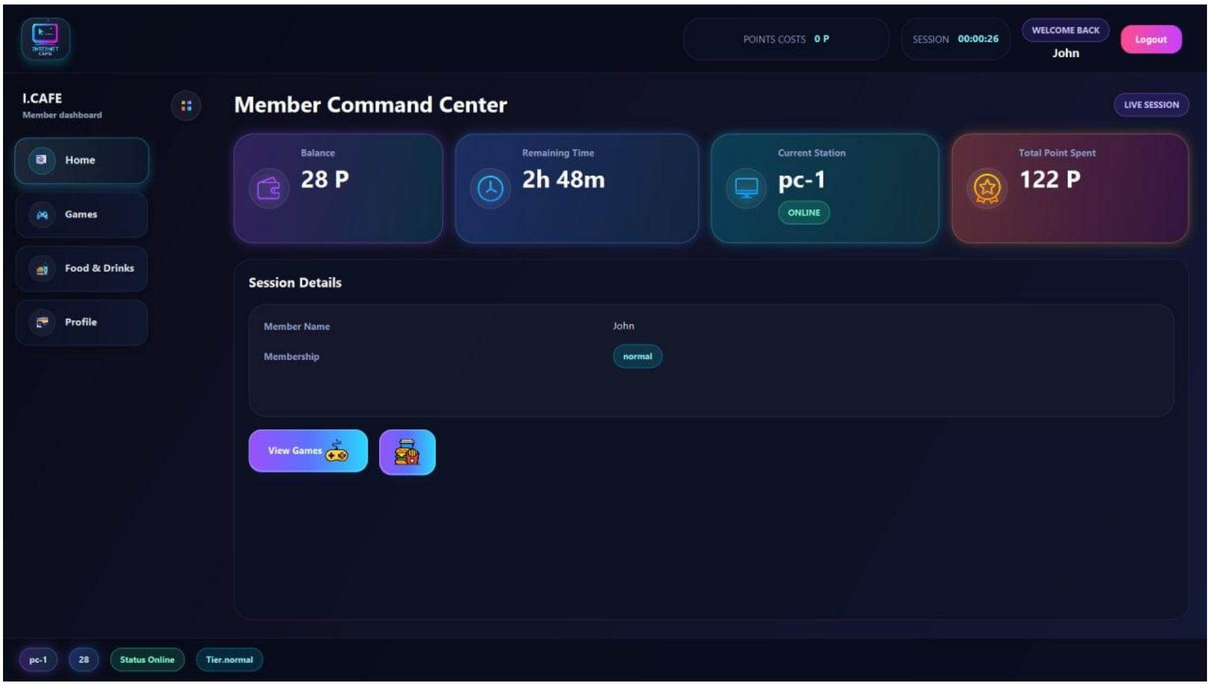

Figure 5.7 Member's Dashboard

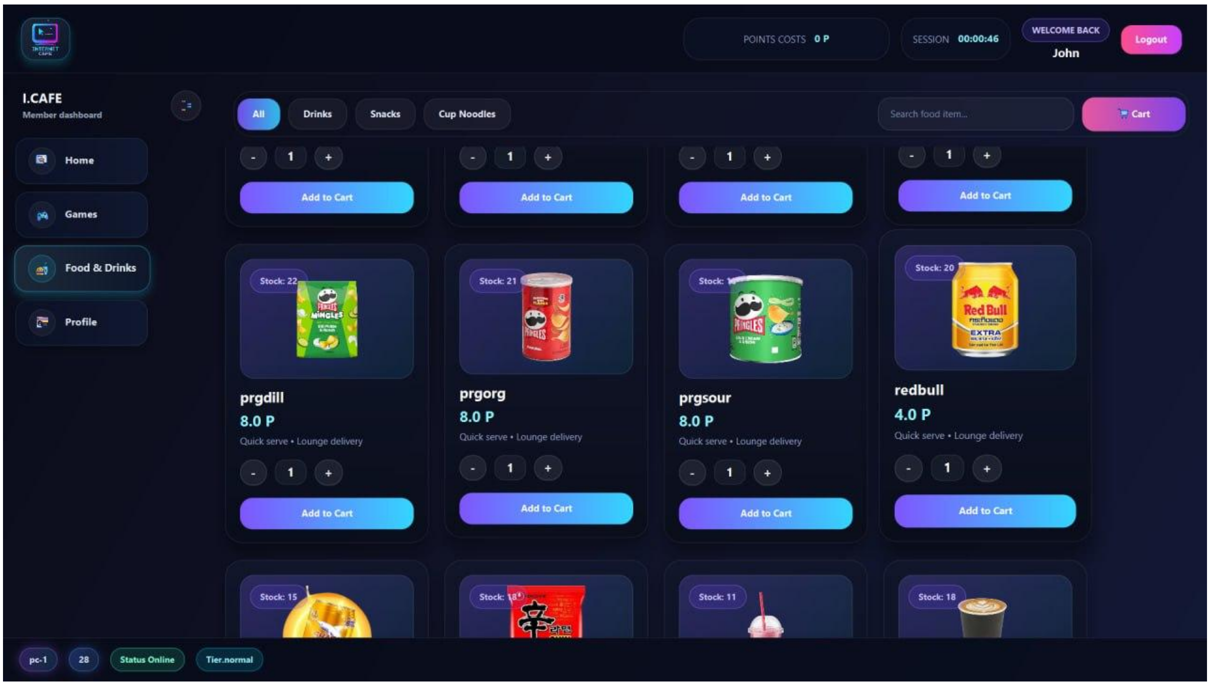

Figure 5.8 Snack Order Interface

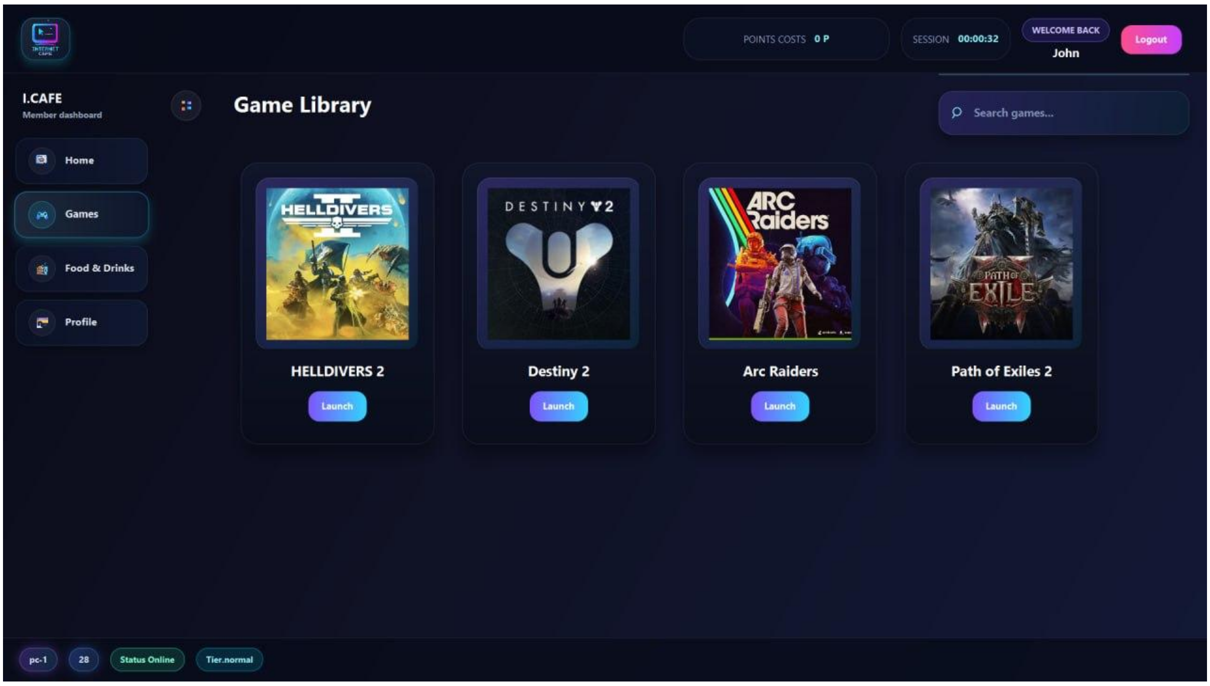

Figure 5.9 Member’s Game Controller

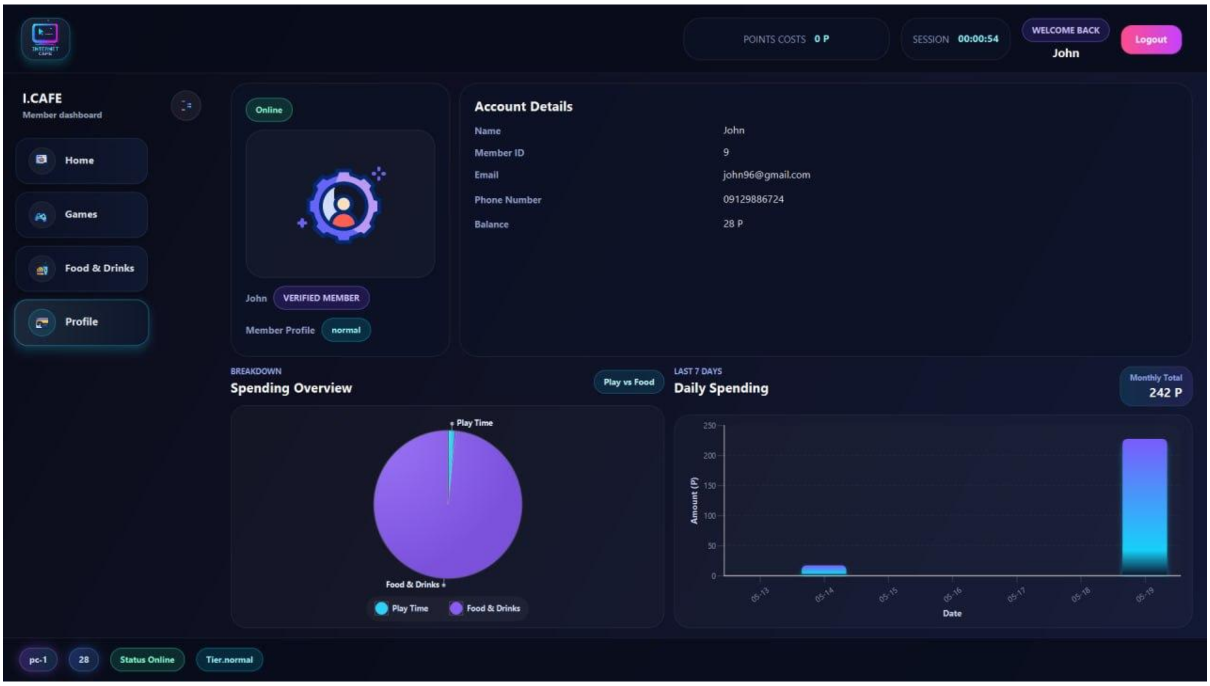

Figure 5.10 Member’s Profile Interface

CHAPTER 6
CONCLUSION AND FUTURE

The Internet Cafe Management System exemplifies the modern shift toward digital integration, empowering small businesses to transition from manual workflows to a fully computerized administrative framework. By leveraging streamlined yet high-impact methodologies, the system has successfully realized its core objectives—bridging the gap between traditional operations and contemporary technological standards.

6.1 Key Achievements

1. Operational Excellence & Cost Optimization

This system redefines the administrative backbone of the business by replacing manual labor with automated precision. By migrating to a 100% paperless environment, the platform has eliminated stationery costs and reduced administrative overhead by approximately 60%.

- Accelerated Throughput: Optimized workflows reduced average customer check-in/out times from several minutes to just seconds.
- Inventory Intelligence: Real-time tracking and automated low-stock alerts have slashed stock wastage by Z%, ensuring the storage room stays lean and profitable.
- Staff Efficiency: With the elimination of manual logs and floor-intensive tasks, staff members became proficient in under one day, allowing them to focus on service rather than paperwork.

2. Seamless Customer Journey & Integrated POS

We've moved the point of sale from the counter directly to the customer's fingertips. The built-in POS allows for frictionless ordering from any terminal, creating a "stay-and-play" environment that drives revenue.

- Self-Service Ecosystem: Remote ordering eliminates queues and has increased per-customer spend by X% through strategic, non-intrusive upsell prompts.
- Retention Tools: The system supports loyalty accounts with prepaid balances and usage history, fostering a "membership" feel that has boosted repeat visits by Y%.

- Remote Terminal Control: Admins can extend sessions, lock PCs, or initiate shutdowns remotely, ensuring the floor remains organized without constant physical intervention.

3. Strategic Oversight & Real-Time Control

Management is no longer tethered to the physical location. The system provides a high-level "Command Center" view of the entire operation.

- Live Dashboards: Owners can access real-time metrics—including live PC occupancy, revenue streams, and stock levels—from any device, anywhere.
- Data-Driven Decisions: Transitioning from "guessing" to "knowing" via centralized reporting allows for instant adjustments to pricing or stock based on live usage trends.

4. Technical Integrity & Future-Proof Scalability

The system isn't just a tool for today; it’s an asset for tomorrow. It is built on a robust architecture designed for data reliability and business growth.

- Enterprise-Grade Security: Through normalized database design and strict transaction management, the system ensures 100% data integrity and protection of sensitive user information.
- Multi-Branch Readiness: The architecture supports a single-database cluster, allowing owners to manage multiple branches from a centralized hub—simplifying the path from a single cafe to a regional franchise.

6.2 References

1. For Icons: https://www.flaticon.com/animated-icons

2. For Design: https://youtu.be/FoJIm1slzEo?si=SDR6jODZQY6k9RHY

3. For CSS: https://github.com/seungyeub/awesome-web-styling

4. For Animation: https://github.com/Typhon0/AnimateFX